Grade 8 Mathematics - Target Content PDF
Topics: Fractions, Algebra, Geometry
For Learning Chat and weak-area remediation

PART 1: FRACTIONS

A fraction represents part of a whole: $a/b$ where $b$ is not 0.

Simplifying: divide numerator and denominator by their GCF.
Example: $12/18 = (12/6)/(18/6) = 2/3$

Adding with unlike denominators: find LCD first.
Example: $1/4 + 2/3 = 3/12 + 8/12 = 11/12$

Multiplying: multiply numerators and denominators.
Example: $2/5 \times 3/4 = 6/20 = 3/10$

Dividing: multiply by the reciprocal.
Example: $3/4 / 1/2 = 3/4 \times 2/1 = 3/2$

Word problem tip: 'half of 2/3' means $1/2 \times 2/3 = 1/3$.

PART 2: ALGEBRA

Linear equations: isolate the variable using inverse operations.
Example: $3x + 7 = 22$
Step 1: $3x = 15$
Step 2: $x = 5$

Quadratic equations have degree 2: $ax^2 + bx + c = 0$.

Square root method: $x^2 - 9 = 0$
$x^2 = 9$, so $x = 3$ or $x = -3$

Factoring: $x^2 + 7x + 12 = (x + 3)(x + 4)$
Set each factor to zero to find roots.

Distributive property: $2(x - 4) = 10$
$2x - 8 = 10$, $2x = 18$, $x = 9$

Discriminant $D = b^2 - 4ac$ tells number of real roots.
For $x^2 + 6x + 9 = 0$: $D = 36 - 36 = 0$ (one repeated root).

PART 3: GEOMETRY

Triangle angle sum: interior angles add to 180 degrees.

Rectangle area = length x width.
Example: 12 cm x 5 cm = 60 $cm^2$

Circle circumference $C = 2 \pi r$.
Example: $r = 7$, $\pi = 22/7$, $C = 44$ cm

Supplementary angles sum to 180 degrees.
If one angle is 65, the other is 115 degrees.

Triangle area = 1/2 x base x height.

Example: base 10, height 6, area = 30 cm^2

Square perimeter = 4 x side.
If perimeter is 36 cm, side = 9 cm.

Study tip: draw a diagram for every geometry word problem.

END OF TARGET CONTENT